Fertility and Housing Tenure Choice: Model Writeup

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1 Model

The model is a discrete-time lifecycle economy with endogenous fertility, housing tenure choice, and spatial sorting. Households save, choose where to live, decide whether to rent or own, and choose fertility. Children enter preferences and resources through consumption and housing requirements. Housing is local, supplied upward sloping in each location, and costly to access because of down-payment and moving frictions. The mechanism is that children require space, space is more expensive in central locations, and tenure constraints affect family formation over the lifecycle.

The environment combines three standard ingredients. The fertility block follows quantitative lifecycle models in which children are chosen by households and affect both current resources and continuation values (Sommer, 2016; Doepke and Kindermann, 2019). The tenure block follows housing-macro models with collateral constraints and discrete owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). In particular, the housing block closely resembles Greaney et al. (2025), a recent paper with focus on space and housing. The spatial block is Rosen-Roback: households sort across locations with different amenities and prices, and local housing supply determines how demand pressure maps into prices (Rosen, 1979; Roback, 1982).

1.1 Environment

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R[1], and die at the end of age J.

There are two residential locations, $\mathcal{I} = \{P, C\}$, where P denotes the periphery and C denotes the center. A unit of housing in location i has purchase price p_i and rental price r_i. Location i also has amenity ξ_i, housing supply scale H_{0i}, and housing supply elasticity η_i. The amenity ξ_i enters the location choice value, H_{0i} shifts the level of local housing supply, and η_i governs the slope of that supply curve. The center is the high-amenity, high-price location in the benchmark.

The individual state at the beginning of a period is $x = (b, d, i, a, n, s)$. Here b is liquid wealth, d is tenure, i is current location, a is age, n is completed fertility, and s is the child-age state.

[1] I might consider simplifying the lifecycle structure to just some stages, to improve computational efficiency. The next step is, indeed, clearly determining what are the empirical targets for the model, and so whether I want to match very well lifecycle profiles or not.

Tenure belongs to $\mathcal{D} \equiv \{R, O_1, \ldots, O_{N_H}\}$. R denotes renting. O_k denotes ownership of a house with physical size H_k, where $H_1 < \cdots < H_{N_H}$. Owners choose from this discrete ladder[2], while renters choose rental space continuously up to a cap $\bar{h}^R$.

1.2 Preferences

Let $m(n, s)$ be the number of dependent children living at home[3]. Completed fertility n and current dependents $m(n, s)$ differ because children eventually leave the household. Flow utility is Stone–Geary over nondurable consumption and housing services, wrapped in CRRA:

$$u(c, \mathbf{h}; n, s) = \frac{\left[(c - \bar{c}(n, s))^{\alpha} \left(\mathbf{h} - \bar{h}(n, s)\right)^{1-\alpha}\right]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} m(n, s), \tag{1}$$

where $\alpha \in (0, 1)$ is the expenditure-share parameter inside the consumption-housing aggregator, $\sigma > 1$ is curvature, and ψ_{child} is the flow value of each dependent child.[4] The Stone–Geary terms $\bar{c}(n, s)$ and $\bar{h}(n, s)$ are minimum consumption and housing needs:

$$\begin{aligned}
\bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n m(n, s), \\
\bar{h}(n, s) &= \bar{h}_0 + \mathbb{I}\{m(n, s) \geq 1\}\bar{h}_{\text{jump}} + \bar{h}_n m(n, s).
\end{aligned}$$

Here $\bar{c}_0$ and $\bar{h}_0$ are baseline needs for a childless household. The parameter $\bar{c}_n$ is the per-child consumption requirement. The parameter $\bar{h}_{\text{jump}}$ is the fixed housing requirement associated with having any dependent child, while $\bar{h}_n$ is the additional housing requirement per dependent child. The first child therefore raises required housing discretely; additional children raise required housing linearly. The local price of space is then part of the cost of children.

Housing services differ by tenure. Let

$$\mathbf{h}(h^O, h^R) = \chi h^O + h^R$$

be the housing services entering utility. Renters have $h^O = 0$ and choose h^R. Owners of rung k have $h^O = H_k$ and $h^R = 0$. The parameter $\chi > 1$ captures the service premium of owner-occupied housing. In market clearing, however, owners demand the physical quantity H_k, not the service quantity χH_k.

At death, households receive warm-glow bequest utility. Let $\psi \in [0, 1)$ denote the proportional transaction cost paid when owner-occupied housing is liquidated. Bequeathable wealth is

$$\widetilde{b} = \begin{cases} b, & d = R, \\ b + (1-\psi) p_i H_k, & d = O_k. \end{cases}$$

The bequest term is

$$\mathcal{B}(\widetilde{b}, n) = \theta_0 (1 + \theta_n n) \frac{(\theta_1 + \max\{\widetilde{b}, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma}. \tag{2}$$

[2] This makes housing more lumpy and makes the model computationally lighter, but might be amended. Some further thinking is required on the housing market structure.

[3] Some notation improvement is probably required, but in any case, the presence of both m and n allows us to distinguish children ever had with children currently at home. which affects the utility.

[4] The position of the ψ_{child} term determines how the child benefit relates to income, so it requires some further thought.

The parameter θ_0 governs the overall strength of the bequest motive. The parameter θ_n scales the bequest motive with completed fertility, so completed fertility affects bequests even after children leave the household. The shift θ_1 controls curvature near zero wealth. The normalization sets bequest utility equal to zero when bequeathable wealth is nonpositive.

1.3 Firms, Income, and Assets

Competitive firms produce the consumption good with aggregate efficiency labor, $Y = AL$. The consumption good is the numeraire. Competition implies $w = A$, where w is the common wage per efficiency unit. Wages do not vary by location, so spatial sorting is driven by amenities, housing costs, tenure frictions, and moving wedges, not by local wage differences.

Workers supply labor inelastically. A worker of age a supplies e_a efficiency units, so labor income is

$$y_a = (1 - \tau_{\text{pay}})we_a, \qquad a \leq J_R,$$

where e_a is the deterministic age-efficiency profile and τ_{pay} is the payroll tax rate. Retired households receive

$$y_a = p^{\text{ret}}, \qquad a > J_R.$$

The pension benefit is determined in equilibrium by the pay-as-you-go government budget.

Given the level distribution G, aggregate efficiency labor supply is

$$L^s(G) = \int \mathbb{I}\{a \leq J_R\} e_a \, dG(x).$$

In equilibrium firms hire this labor supply, so $L = L^s(G)$ and output is $Y = AL$.

Liquid wealth earns exogeneous gross return $R_b \equiv 1 + q$. A renter cannot borrow:

$$b' \geq 0.$$

At origination, a buyer of house H_k in location i must satisfy

$$b \geq -\phi p_i H_k, \tag{3}$$

where b denotes liquid wealth immediately after the purchase. Here ϕ is the financed share of house value. The required down payment is therefore $(1 - \phi)p_i H_k$. A homeowner who keeps the current house is not forced to delever after a price change, but cannot increase debt while below the origination bound:

$$b' \geq \underline{b}_i^O(b, H_k) \equiv \min\{b, -\phi p_i H_k\}.$$

1.4 Housing and Budgets

Given rental price r_i, a renter chooses c, h^R, and b' subject to

$$c + r_i h^R + b' = R_b b + y_a, \qquad 0 \leq h^R \leq \bar{h}^R, \qquad b' \geq 0.$$

An owner of house H_k chooses c and b' subject to

$$c + \delta p_i H_k + b' = R_b b + y_a, \qquad b' \geq \underline{b}_i^O(b, H_k).$$

Mortgage debt is negative liquid wealth, so mortgage interest is already embedded in $R_b b$. Owners pay maintenance directly.

Housing adjustment is costly. The same transaction-cost parameter ψ used in bequeathable wealth applies when a household sells a house. Selling a house in location i yields net proceeds

$$(1 - \psi) p_i H_k.$$

A moving owner sells the origin house before choosing tenure in the destination. A stayer can keep the current house, sell and rent, or move to another owner rung if the budget and collateral constraints allow it.

1.5 Fertility and Child Aging

During the fertile window

$$a \in \{A_f^{\text{start}}, \ldots, A_f^{\text{end}}\},$$

a childless household chooses $n' \in \mathcal{N} \equiv \{0, 1, \ldots, \bar{n}\}$. The choice $n' = 0$ means no birth in the current period. A positive choice $n' > 0$ fixes completed fertility forever. A household that reaches the end of the fertile window without a positive fertility choice remains childless. Children are born together and age together through K child stages.[5]

Let $s = 0$ denote no dependent children and $s = 1, \ldots, K$ denote dependent child stages. At the moment of a positive fertility choice, the household moves into state $s = 1$. After that, child aging is stochastic but sequential. For $s < K$, children either remain in stage s or move to stage $s + 1$. At stage K, they either remain dependent or mature and leave the household, in which case the household returns to $s = 0$. No other child-age transitions have positive probability. The matrix Π^c collects these transition probabilities. In the parameterization, dependent childhood lasts about 18 years. While children are at home, $m(n, s) = n$. After maturity, $m(n, s) = 0$, so the consumption and housing floors return to their baseline levels, while completed fertility remains relevant through the bequest motive. The mature city-born flow is

$$B(G) = \int n \Pi^c(0 \mid K, n) \mathbb{I}\{s = K\} dG(x). \tag{4}$$

Mature city-born children are potential entrants; the outside option determines whether they enter the modeled city.

1.6 Household Problem

Within the period, the order of decisions is fertility, location, tenure and house size, and then consumption, rental size, and savings. The discount factor is β. Let the terminal value be

$$V_{J+1}(b, d, i, n, s) = \mathcal{B}(\widetilde{b}, n),$$

[5] The model therefore can't capture second-birth effects and planning. This is to simplify state space. One could consider other alterntives, but would need to simplify on other channels.

and define the continuation value after child aging by

$$\bar{V}_{a+1}(b', d', i', n, s) = \sum_{s'} \Pi^c(s'|s, n) V_{a+1}(b', d', i', n, s').$$

Conditional on being a renter in location i, the household value is

$$\begin{aligned} V_a^R(b, i, n, s) = \max_{c, h^R, b'} \quad & u(c, \mathbf{h}(0, h^R); n, s) + \beta \bar{V}_{a+1}(b', R, i, n, s) \\ \text{s.t.} \quad & c + r_i h^R + b' = R_b b + y_a, \\ & 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0. \end{aligned} \tag{5}$$

Conditional on owning house H_k in location i, the value is

$$\begin{aligned} V_a^{O_k}(b, i, n, s) = \max_{c, b'} \quad & u(c, \mathbf{h}(H_k, 0); n, s) + \beta \bar{V}_{a+1}(b', O_k, i, n, s) \\ \text{s.t.} \quad & c + \delta p_i H_k + b' = R_b b + y_a, \\ & b' \ge \underline{b}_i^O(b, H_k). \end{aligned} \tag{6}$$

Define $H(R) = 0$ and $H(O_k) = H_k$. After choosing destination j, a household with state (b, d, i) carries liquid wealth and owner-occupied housing

$$\widetilde{b}^R(b, d, i; j) = b + \mathbb{I}\{j \ne i\}(1 - \psi) p_i H(d), \qquad \widetilde{H}^R(d, i; j) = \mathbb{I}\{j = i\} H(d).$$

If it then chooses tenure d' in destination j, liquid wealth is

$$\widetilde{b}^H(b, d, i; j, d') = \widetilde{b}^R(b, d, i; j) + \mathbb{I}\{H(d') \ne \widetilde{H}^R(d, i; j)\} \left[(1 - \psi) p_j \widetilde{H}^R(d, i; j) - p_j H(d')\right]. \tag{7}$$

The feasible tenure set in destination j is

$$\mathcal{D}(b, d, i; j) = \left\{ d' \in \mathcal{D} : \big(j = i \text{ and } d' = d\big) \text{ or } \widetilde{b}^H(b, d, i; j, d') \ge \underline{b}_j(d') \right\},$$

where $\underline{b}_j(R) = 0$ and $\underline{b}_j(O_k) = -\phi p_j H_k$. The tenure value in destination j is

$$V_a^T(b, d, i; j, n, s) = \max_{d' \in \mathcal{D}(b, d, i; j)} V_a^{d'}\left(\widetilde{b}^H(b, d, i; j, d'), j, n, s\right). \tag{8}$$

Here $V_a^{d'} = V_a^R$ if $d' = R$ and $V_a^{d'} = V_a^{O_k}$ if $d' = O_k$.

Location choice has additive Type-I extreme-value shocks with scale κ_ℓ. The parameter κ_ℓ governs the amount of residual dispersion in location choices; as κ_ℓ goes to zero, location choice converges to a deterministic maximum. A household currently in location i evaluates destination j using

$$\widetilde{V}_a^L(j \mid \cdot) = V_a^T(b, d, i; j, n, s) + \xi_j - \zeta_{ij},$$

where ζ_{ij} is the bilateral moving wedge, with ζ_{ii} normalized to zero. The destination probabilities are

$$\pi_a^L(j \mid \cdot) = \frac{\exp\left(\widetilde{V}_a^L(j \mid \cdot)/\kappa_\ell\right)}{\sum_{m \in \mathcal{I}} \exp\left(\widetilde{V}_a^L(m \mid \cdot)/\kappa_\ell\right)}, \tag{9}$$

and the location value is

$$V_a^L(b,d,i,n,s) = \kappa_\ell \log \sum_{m\in\mathcal{I}} \exp\left(\widetilde{V}_a^L(m \mid \cdot)/\kappa_\ell\right). \tag{10}$$

Childless households in the fertile window face a stopping problem with additive Type-I extreme-value shocks with scale κ_f. The option $n' = 0$ means delaying fertility for one period. A positive option $n' > 0$ fixes completed fertility permanently. Let $s(n') = 0$ if $n' = 0$ and $s(n') = 1$ if $n' > 0$. The value of choosing option $n' \in \mathcal{N}$ is

$$\widetilde{V}_a^F(n' \mid \cdot) = V_a^L(b,d,i,n',s(n')).$$

Fertility probabilities are

$$\pi_a^F(n' \mid \cdot) = \frac{\exp\left(\widetilde{V}_a^F(n' \mid \cdot)/\kappa_f\right)}{\sum_{m\in\mathcal{N}} \exp\left(\widetilde{V}_a^F(m \mid \cdot)/\kappa_f\right)}, \tag{11}$$

and the full household value is

$$V_a(b,d,i,n,s) = \kappa_f \log \sum_{m\in\mathcal{N}} \exp\left(\widetilde{V}_a^F(m \mid \cdot)/\kappa_f\right). \tag{12}$$

Outside the fertility window, or after fertility has already been chosen, this stage is degenerate and $V_a(b,d,i,n,s) = V_a^L(b,d,i,n,s)$.

1.7 Entry, Outside Option, and City Size

Endogenous fertility changes the flow of children who mature into adult households. The model therefore needs a rule for which potential young adults enter the modeled city and a rule for the scale of the city. The benchmark rule is an outside-option entry problem for potential entrants.

At the beginning of adult life, a potential entrant is either a mature city-born child or an outside-origin potential immigrant. If the entrant chooses modeled location $i \in \mathcal{I}$, she begins adult life as a childless renter with liquid wealth b^E:

$$x_i^E = (b^E, R, i, 1, 0, 0), \qquad W_i^E(p) = V_1(b^E, R, i, 1, 0, 0; p).$$

The entrant can instead choose the outside economy, which has value $\bar{W}^E$. Potential entrants draw Type-I extreme-value tastes over the modeled locations and the outside economy, with scale parameter κ_E. The unconditional probability of entering modeled location i is

$$\pi_i^E(p) = \frac{\exp\left(W_i^E(p)/\kappa_E\right)}{\exp\left(\bar{W}^E/\kappa_E\right) + \sum_{m\in\mathcal{I}} \exp\left(W_m^E(p)/\kappa_E\right)}, \qquad i \in \mathcal{I}, \tag{13}$$

and the probability of choosing the outside economy is

$$\pi_{\text{out}}^E(p) = \frac{\exp\left(\bar{W}^E/\kappa_E\right)}{\exp\left(\bar{W}^E/\kappa_E\right) + \sum_{m\in\mathcal{I}} \exp\left(W_m^E(p)/\kappa_E\right)}. \tag{14}$$

The total probability of entering the modeled city is

$$q^E(p) = \sum_{i \in \mathcal{I}} \pi_i^E(p) = 1 - \pi_{\text{out}}^E(p). \tag{15}$$

Thus $\pi_i^E(p)$ determines where entrants go inside the modeled city, while $q^E(p)$ determines how many potential entrants enter the city at all. The entry shocks governed by κ_E are distinct from the incumbent location shocks governed by κ_ℓ.

Let G denote the level distribution of adult households in the modeled city, and let

$$S = \int dG.$$

Write the level distribution as

$$G = Sg, \qquad \int dg = 1,$$

where g is the normalized composition of the city. This convention separates shares from levels: g gives the composition of adult households, S gives total adult mass, and $G = Sg$ gives actual masses.

Each period the pool of potential young-adult entrants is the sum of an exogenous outside-born flow $M \geq 0$ and the mature city-born flow $B(G)$:

$$Q(G) = M + B(G). \tag{16}$$

Entry into location i is

$$E_i(G; p) = Q(G)\pi_i^E(p). \tag{17}$$

The remaining mass $Q(G)\pi_{\text{out}}^E(p) = Q(G)(1 - q^E(p))$ stays in the outside economy.

For a candidate price vector p, household policies imply a normalized stationary composition $g(p)$. Let

$$E_0(p) = \int \mathbb{I}\{a = 1\} dg(p)$$

be the replacement inflow of young adults required to reproduce one unit of the stationary composition. With deterministic death at the end of the lifecycle, this is also the per-unit mass of adults exiting through death in stationarity. Let $B_0(p) = B(g(p))$ be the mature city-born flow per unit of city scale. The scalar scale condition equates required young-adult replacement to actual city entry:

$$SE_0(p) = q^E(p)\left[M + SB_0(p)\right]. \tag{18}$$

If

$$E_0(p) > q^E(p)B_0(p), \tag{19}$$

then the stationary city scale is

$$S(p) = \frac{q^E(p)M}{E_0(p) - q^E(p)B_0(p)}. \tag{20}$$

The denominator is the net replacement gap. If condition (19) fails, each unit of city scale produces enough city-entering mature children to weakly cover its own replacement requirement, and the maintained stationary closure does not admit a finite city scale.

The baseline uses the normalization $S^* = 1$. The normalized benchmark solution implies E_0^* and B_0^*. The outside value $\bar{W}^E$ is chosen to generate the benchmark city-entry probability $q^{E,*}$. Given those objects, the outside-born potential entrant mass is set residually from (18):

$$M = \frac{E_0^*}{q^{E,*}} - B_0^*.$$

Thus E_0^*, B_0^*, and $q^{E,*}$ are not three independent free objects: the first two come from the normalized benchmark composition, the third pins down the outside margin, and M is residual. In counterfactuals, M, $\bar{W}^E$, and κ_E are held fixed, while (20) determines the new scale $S(p)$. Location-specific stationarity is handled by the vector of entry masses $\{E_i(G;p)\}_{i\in\mathcal{I}}$ in (17); the scalar equation only pins down total city scale.

1.8 Housing Supply and Market Clearing

Housing supply in location i is

$$H_i^S(r_i) = H_{0i} r_i^{\eta_i}. \tag{21}$$

The scale H_{0i} captures reduced-form land costs, regulation, and other supply-side wedges. The elasticity η_i governs the construction response. Thus the supply side has only a level parameter and an elasticity in each location.

Writing ℓ for the household's current location, aggregate physical housing demand in location i is

$$H_i^D = \int \left[\mathbb{I}\{\ell = i, d = R\} h^R(b,d,\ell,a,n,s) + \sum_{k=1}^{N_H} \mathbb{I}\{\ell = i, d = O_k\} H_k \right] dG(b,d,\ell,a,n,s), \tag{22}$$

House prices satisfy the user-cost relation

$$r_i = (q+\delta)p_i.$$

2 Stationary Equilibrium

The equilibrium is stationary in prices, policies, and the level distribution of adult households. The level distribution is denoted by G; its total mass is $S = \int dG$, and its normalized composition is $g = G/S$. Entry, aging, fertility, location choice, tenure choice, savings, and death reproduce the same level distribution every period.

Definition 1 (Stationary equilibrium)**.** *A stationary equilibrium is a collection*

$$\begin{aligned}\mathcal{X}^* = \big(&\{p_i, r_i\}_{i\in\mathcal{I}}, w, L, Y, p^{\text{ret}}, \{V_a, V_a^R, V_a^T, V_a^L, \widetilde{V}_a^L, \widetilde{V}_a^F\}_a, \{V_a^{O_k}\}_{a,k}, \{c, h^R, b', d', i', n'\},\\ &\{\pi^L, \pi^F, \pi^E, \pi^E_{\text{out}}, q^E\}, G, S, g, \{E_i, H_i^D, H_i^S\}_{i\in\mathcal{I}}\big)\end{aligned}$$

such that:

1. ***Household optimization.*** *Given prices, amenities, and moving wedges, value functions solve the household problem in (5)–(6), with tenure choice (8), location probabilities (9), fertility probabilities (11), and terminal bequest utility (2). The policies c, h^R, b', d', i', and n' are associated optimal policies.*

2. ***Entry and outside option.*** *The child-aging matrix implies a mature city-born flow $B(G)$. The potential entrant pool is $Q(G) = M + B(G)$. Entry probabilities are $\{\pi_i^E(p)\}_{i\in\mathcal{I}}$ and $\pi_{\text{out}}^E(p)$, the city entry rate is $q^E(p) = 1 - \pi_{\text{out}}^E(p) = \sum_i \pi_i^E(p)$, and entry masses satisfy*

$$E_i(G;p) = Q(G)\pi_i^E(p), \qquad i \in \mathcal{I}.$$

The outside mass is $Q(G)(1 - q^E(p))$.

3. ***Firms and labor market.*** *The common wage satisfies $w = A$. Aggregate efficiency labor supply is*

$$L^s(G) = \int \mathbb{I}\{a \leq J_R\} e_a \, dG(x),$$

and the labor market clears:

$$L = L^s(G).$$

Output is $Y = AL$.

4. ***Housing markets and user cost.*** *In every location,*

$$H_i^D = H_i^S(r_i), \qquad r_i = (q + \delta)p_i.$$

5. ***Government budget.*** *Payroll-tax revenue finances pension benefits:*

$$\tau_{\text{pay}} wL = p^{\text{ret}} \sum_i \sum_{a>J_R} G_i(a).$$

Here $G_i(a)$ is the marginal mass of households of age a living in location i.

6. ***Stationary distribution.*** *The distribution G is the level distribution of adult households living in the modeled city. For any measurable set $\mathcal{A}$ of adult states,*

$$G'(\mathcal{A}) = \int \mathcal{P}(\mathcal{A} \mid x; p) \, dG(x) + \sum_{i\in\mathcal{I}} E_i(G;p)\mathbb{I}\{x_i^E \in \mathcal{A}\},$$

where $\mathcal{P}(\cdot \mid x; p)$ is the transition induced by optimal policies, fertility and location probabilities, child aging, deterministic aging, and death for incumbent households. Death puts no mass on the next-period adult city state space. Mature city-born children enter the adult city distribution only through the entry masses $\{E_i(G;p)\}_{i\in\mathcal{I}}$; the remaining potential entrants choose the outside economy and are not part of G'. Stationarity requires $G' = G$. Entrants enter at $x_i^E = (b^E, R, i, 1, 0, 0)$. Total adult population is $S = \int dG(x)$ and is an equilibrium outcome.

3 Population Closure: Alternatives

The proposed solution above, with an outside option entry, does not achieve much more than a pure normalization. The main gain is that we can get a scale effect in counterfactuals. I did not fully get how to get a scale effect in the steady state. Here i briefly sum up the alternatives and thoughts.

Fixed adult mass. The simplest closure (as presented in April) has $S = 1$ and keeps adult population fixed. Fertility then changes the composition of households, but not the mass of households in the city. Housing demand changes because parents want more space, sort differently, and choose tenure differently. It does not change because the city becomes larger. This is close in spirit to exercises that take the adult demographic envelope or prices as given, including Moreno-Maldonado and Santamaria (2024) and Karabarbounis and Nord (2025). The advantage is tractability; the cost is that the scale channel is shut down.

Outside-option entry. The benchmark uses the outside-option version. It keeps the same stationary accounting logic, but makes city entry depend on the value of entering the city relative to an outside economy:

$$SE_0(p) = q^E(p)\left[M + SB_0(p)\right].$$

The equilibrium is still stationary: it solves for a level S, not for a population growth rate. Relative to the reduced-form valve, housing prices discipline city size through entry values. If fertility raises the flow of mature city-born children, the potential-entrant pool rises. If the larger city raises housing prices, entry values fall and the outside option absorbs more of the potential flow. The model then has a scale channel without becoming a theory of city growth.

Reduced-form renewal valve. This is the reduced version of the benchmark closure. It keeps the same renewal accounting, but replaces the value-based city-entry probability $q^E(p)$ with a fixed retention parameter ρ:

$$SE_0(p) = M + \rho SB_0(p), \qquad S = \frac{M}{E_0(p) - \rho B_0(p)}.$$

The city receives an outside flow M and retains a fixed fraction ρ of mature city-born children. The left-hand side is the young-adult entry flow required to keep a city of size S stationary. The right-hand side is the flow that actually enters: outside-born entrants plus the retained city-born flow. If fertility raises $B_0(p)$, the stationary scale S rises and aggregate housing demand rises through $H_i^D = S\widehat{H}_i^D$. This delivers a scale channel without a transition path or balanced growth. What is lost, relative to the benchmark, is the entry response to city values. Housing prices can still affect fertility and housing choices inside the household problem, but they no longer directly change the fraction of potential entrants who choose the modeled city.

Full demographic city growth. The most structural alternative is a sequence model in which births today become adult households in future periods, the age distribution evolves over calendar time, and housing markets clear along the transition. This is the natural object for a paper about city growth and fertility dynamics, and it is the direction taken by Coeurdacier et al. (2023). It requires a path for age masses, migration flows, survival, and housing supply. The object here is instead a stationary city in which fertility, housing demand, tenure, and sorting are jointly determined. Aside from the computational challenge, the problem here is also that it's much harder to decide what to match and target, what is the desired outlook and how to do counterfactuals.

Why not BGP? One suggestion made in the talk was to just have a BGP. The way I see it, this is not the correct thing to do. First, with the spatial dimension it creates numerous problems, discussed in the January committee meeting. But aside from that, BGP trades off here one simplification with another: it would allow objects to respond endogeneously to scale, but at the cost of 'normalizing' the population dynamics. That is, in a BGP, the population distribution follows a travelling wave, so everything stays constant in the age distribution (so if today there is twice the amount of 35 year olds as 50 year olds, the same is tomorrow, even if the absolute number is lower or higher). In a BGP with TFR=1.7, for example, with one couple having less then two kids, the population would be shrinking at a constant rate, with the age masses constant. This is also counterfactual: today, we have negative TFR, but positive population growth, due to past fertility rates. The only way to actually match this is with the solution in the paragraph below, full demographic city growth, which is a much more complex object. So, if one wants to avoid that, it seems to me like I need to choose what kind of normalization I want. I might definitely be missing something here, so I am happy to discuss this further.

A Outside-Option Entry Algorithm

This appendix restates the benchmark closure as an algorithm. The key distinction is that the benchmark uses the outside margin to normalize the level of the economy, while counterfactuals hold the outside environment fixed and allow city scale to move.

1. **Solve the normalized benchmark.** For a candidate parameter vector and benchmark prices, solve the household problem and the stationary normalized composition g^* with $\int dg^* = 1$.

2. **Compute per-unit demographic flows.** From g^*, compute the replacement inflow E_0^* and the mature city-born flow $B_0^* = B(g^*)$. These are per unit of city scale, not aggregate masses.

3. **Choose the outside value.** Given entry values $\{W_i^E(p^*)\}_{i\in\mathcal{I}}$, choose $\bar{W}^E$ so that the model matches the benchmark city-entry probability $q^{E,*}$. Equivalently, this pins down the benchmark outside margin.

4. **Set the outside-origin potential entrant mass.** Normalize benchmark city scale to $S^* = 1$ and set

   $$M = \frac{E_0^*}{q^{E,*}} - B_0^*.$$

   Thus M is residual: it is the outside-origin potential entrant mass required to make the normalized benchmark stationary.

5. **Run counterfactuals.** Hold M, $\bar{W}^E$, and κ_E fixed. For each counterfactual price vector p, recompute household behavior, $E_0(p)$, $B_0(p)$, and $q^E(p)$, and set city scale using

   $$S(p) = \frac{q^E(p)M}{E_0(p) - q^E(p)B_0(p)}.$$

The scalar equation determines total scale. The vector of entry probabilities $\{\pi_i^E(p)\}_{i\in\mathcal{I}}$ determines how entrant mass is allocated across locations, which is required for location-specific stationarity.

References

Coeurdacier, N., Combes, P.-P., Gobillon, L., and Oswald, F. (2023). Fertility, housing costs and city growth. Unpublished manuscript.

Doepke, M. and Kindermann, F. (2019). Bargaining over Babies: Theory, Evidence, and Policy Implications. *American Economic Review*, 109(9):3264–3306.

Greaney, B., Parkhomenko, A., and Van Nieuwerburgh, S. (2025). Dynamic Urban Economics. *NBER Working Paper*.

Henderson, J. V. and Ioannides, Y. M. (1983). A model of housing tenure choice. *American Economic Review*, 73(1):98–113.

Karabarbounis, L. and Nord, L. (2025). Fertility choices and the cost of living. Working Paper, University of Minnesota and University of Pennsylvania. Presented at SED 2025, Copenhagen.

Moreno-Maldonado, A. and Santamaria, C. (2024). Delayed childbearing and urban revival: A structural approach. Working paper.

Roback, J. (1982). Wages, rents, and the quality of life. *Journal of Political Economy*, 90(6):1257–1278.

Rosen, S. (1979). Wage-based indexes of urban quality of life. In *Current Issues in Urban Economics*, pages 74–104. Johns Hopkins University Press.

Sommer, K. (2016). Fertility choice in a life cycle model with idiosyncratic uninsurable earnings risk. *Journal of Monetary Economics*, 83:27–38.

Sommer, K. and Sullivan, P. (2018). Implications of US tax policy for house prices, rents, and homeownership. *American Economic Review*, 108(2):241–274.

Sommer, K., Sullivan, P., and Verbrugge, R. (2013). The equilibrium effect of fundamentals on house prices and rents. *Journal of Monetary Economics*, 60(7):854–870.